



Full paper/Mémoire

# Design and synthesis of biobased epoxy thermosets from biorenewable resources



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## ABSTRACT

Biobased diepoxy synthons derived from isoeugenol, eugenol or resorcinol (DGE-isoEu, DGE-Eu and DGER, respectively) have been used as epoxy monomers in replacement of the diglycidyl ether of bisphenol A (DGEBA). Their curing with six different biobased anhydride hardeners leads to fully biobased epoxy thermosets. These materials exhibit interesting thermal and mechanical properties comparable to those obtained with conventional petrosourced DGEBA-based epoxy resins cured in similar conditions. In particular, a high  $T_g$  in the range of 90–130 °C and instantaneous moduli higher than 4.3 GPa have been recorded. These good performances are very encouraging, making these new fully biobased epoxy thermosets compatible with the usual structural application of epoxy materials.

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## 1. Introduction

Thermoset epoxy resins are used in a wide variety of applications, for example, composites, aeronautic and automotive components [1]. Indeed, they have high performances among the available commercial resins. They exhibit excellent adhesion and mechanical properties as well as chemical, electrical and heat resistance. Currently, most of the epoxy resins and hardeners used are derived from oil resources leading to the harmful environmental impact. Moreover, bisphenol A is widely used as a basic

building block for the preparation of these epoxy resins, in particular, the well-known diglycidyl ether of bisphenol A (DGEBA). However, bisphenol A was recently proved to be an endocrine-disrupting chemical [2,3] and a water contaminant [4]. Therefore, its substitution with biobased building blocks is strongly encouraged, which would also reduce the petrochemical contribution [5].

During the last decades, many studies were devoted to develop a wide variety of molecules for the preparation of sustainable epoxy resins [6,7]. A part of this work concerns the replacement of DGEBA in epoxy resins. The authors were obviously interested in synthesizing diepoxy components containing rigid aromatic rings to obtain polymers with high modulus and high glass transition temperatures. For this purpose, Auvergne et al. [8] and also Ding and Matharu [9] reviewed the potential of polyphenols and

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tannins. More recently, Aouf et al. [10,11], Yang et al. [12] and Shibata and Ohkita [13] reported biobased diepoxy resins obtained, respectively, from functionalization of gallic and vanillic acids, plant-based phenolic acids or vanillin derivatives. An alternative way to produce eugenol-based epoxy resins was proposed by Qin et al. [14] and Wan et al. [15,16]. Other appropriate candidates are molecules extracted from lignin, which are very competitive to DGEBA-based thermosets as reviewed by Llevot et al. [17] and Ferdosian et al. [18]. Recently, such a lignin-based epoxy resin was reported for potential application in epoxy asphalt [19]. Thus, lignin-based epoxy resins are currently the subject of innovative and intense investigations [20].

In this context, we recently described a suitable lignin synthon for epoxy thermoset applications, 2-[3-methoxy-4-(2-oxiranylmethoxy)phenyl]-3-methyloxirane, named DGE-isoEu [21]. This new biobased monomer was prepared according to a two-step procedure, from isoeugenol, catalytically fragmented from lignin. Diepoxy derivative of isoeugenol (DGE-isoEu) was cured satisfactorily with acid derivative hardeners to obtain thermosetting epoxy polymer having high  $T_g$  and promising mechanical properties. Continuing our approach and with the aim to develop fully biobased epoxy thermoset materials, the curing of the DGE-isoEu with biobased anhydride hardeners is studied in this article. Compared with the classical diamine hardeners, these curing agents present several major advantages. First, they avoid the use of toxic aliphatic diamine. Moreover, as reported by Paramarta and Webster [22], anhydride hardeners are preferred to diamines for their higher reactivity, especially in the case of internal DGE-isoEu epoxide that is less reactive as compared to terminal epoxide. Furthermore, to reduce the environmental impacts of the epoxy thermosets, the curing agents used in this study are synthesized with high yield from biorenewable and inexpensive dicarboxylic acid agents [23]. Recently an innovative route to renewable anhydrides from biobased furanics has also been described by Thiyagarajan et al. [24]. It is worth noting that recently much attention has also been paid to biobased amine hardeners as reviewed by Froidevaux et al. [25] and Fache et al. [26]. It is also noticeable that, after curing, some of the tested hardeners still contain a rigid cycle: this is anticipated to produce epoxy thermosets with high glass temperature and high modulus.

In this article, DGE-isoEu or diepoxy derivative of eugenol (DGE-Eu) was successively cured with six biobased anhydride hardeners: the resulting materials were analysed using differential scanning calorimetry (DSC), thermogravimetric analysis (TGA), Fourier transform infrared spectroscopy (FT-IR) and nanoindentation. These results were compared with those obtained by curing commercially available epoxy resins (DGEBA and the biobased diglycidyl ether resorcinol [DGER] resin) with six anhydride hardeners.

## 2. Experimental section

### 2.1. Materials

DGE-isoEu was prepared from biobased isoeugenol [27] and biobased epichlorohydrin [28] according to our

previously published procedure [21]. DGE-Eu was obtained as recently reported by Qin et al. [14]. XRD single-crystal analysis of DGE-Eu was recently resolved by some of us [29]. DGEBA and DGER were purchased from Sigma–Aldrich.

Six anhydride curing agents were synthesized according to Ref. [23]: diphenic anhydride (DPA), glutaric anhydride (GA), phthalic anhydride (PA), camphoric anhydride (CA), succinic anhydride (SA) and itaconic anhydride (IA). Anhydrides were used after the purification step (separation from  $MgCl_2$  by sublimation). 1,2-Dimethylimidazole (DMID, 95% purity) was purchased from Sigma–Aldrich and was used without any further purification.

Molecular representations of reagents used in the resin manufacturing process are presented in Figs. 1 and 2.

### 2.2. General methods and instrumentation

#### 2.2.1. Fourier transform infrared spectroscopy

FT-IR spectra were recorded using a Bruker Vector 22 fitted with a specac MKII Golden Gate Diamond Attenuated Total Reflectance device equipped with ZnSe lenses in the 4000–500  $cm^{-1}$  wavenumber range (12 scans at spectral resolution of 4  $cm^{-1}$ ).

#### 2.2.2. Differential scanning calorimetry

DSC analyses were carried out using TA MDSC 2920 under nitrogen flow (60  $mL\ min^{-1}$ ) with a sample mass of  $10 \pm 3$  mg. To study curing reaction, samples were heated from  $-20$  to  $180\ ^\circ C$  at a heating rate of  $5\ ^\circ C\ min^{-1}$ . To determine glass transition temperature ( $T_g$ ), the cured resin was heated from  $-20$  to  $180\ ^\circ C$  at a heating rate of  $20\ ^\circ C\ min^{-1}$ .

#### 2.2.3. Thermogravimetric analyses

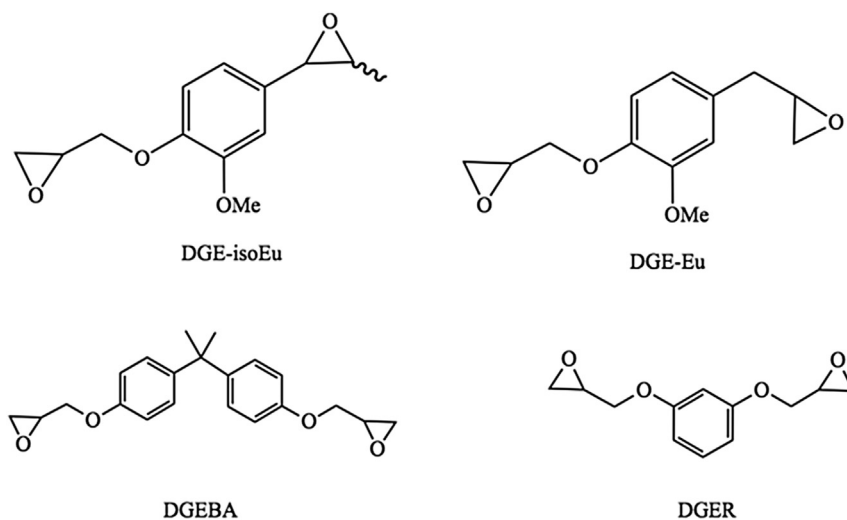
Thermogravimetric analyses were performed using a TA Instruments TGA Q600 thermoanalyser using aluminium pans. Samples (5–10 mg) were heated from room temperature to  $800\ ^\circ C$  at a rate of  $20\ ^\circ C\ min^{-1}$  under air flow (100  $mL\ min^{-1}$ ). Weight loss percentages were determined using the TA Universal Analysis 2000 software accompanying the instrument.

#### 2.2.4. Nanoindentation mechanical property analyses

Nanoindentation experiments were performed using an ultra nanoindenter from Anton Paar. A Berkovich indenter was used. The area function was calibrated by indenting in a fused silica sample with a known Young's modulus and Poisson's ratio. Before carrying out the tests, the samples were polished with SiC paper, alumina particles until 0.03  $\mu m$  and then colloidal silica. The instantaneous indentation modulus  $E_i$  and hardness  $H$  at different indentation depths, the relaxed indentation modulus  $E_R$  and finally the strain rate sensitivity  $m$  were determined from multicycle indentations, indentation at different constant strain rate  $\dot{h}/h$  (where  $h$  is the indentation depth) and trapezoidal holding force tests with linear loading and unloading segments. The indentation protocols were the same as used previously [21].

#### 2.2.5. Epoxy resin curing protocol

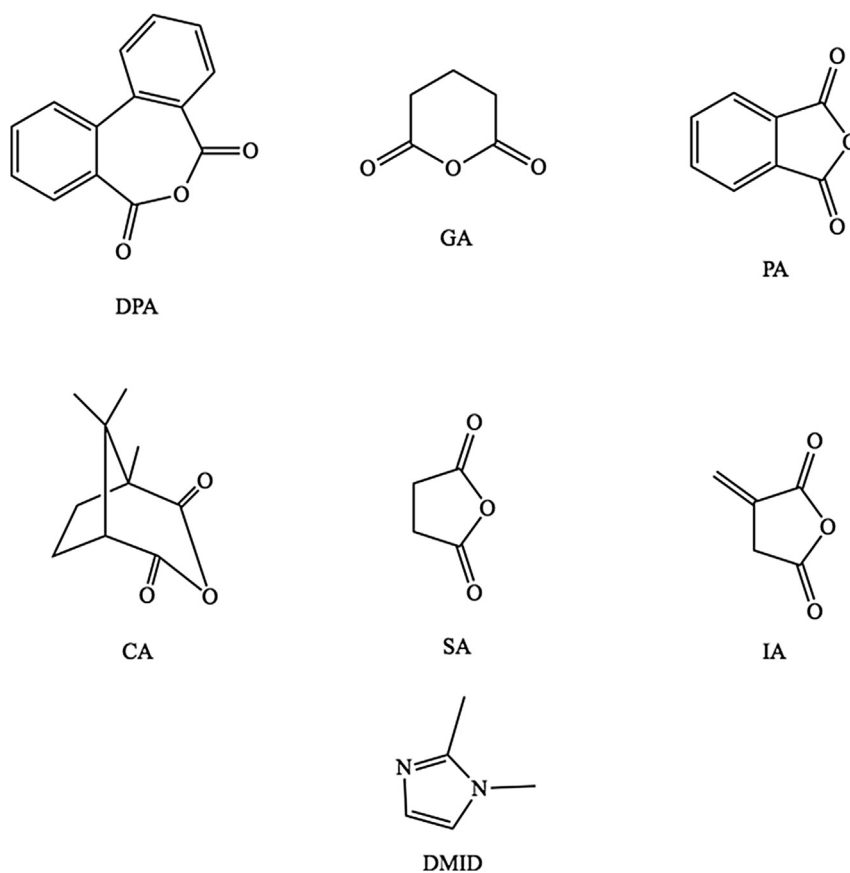
A comparable protocol as described in our previous work [21] was carried out for each sample of this study. Epoxy



**Fig. 1.** Molecular representations of reagents used in the resin manufacturing process. Epoxy monomers: diepoxy derivative of isoeugenol (DGE-isoEu), diepoxy derivative of eugenol (DGE-Eu), diglycidyl ether of bisphenol A (DGEBA), diglycidyl ether of resorcinol (DGER).

monomers and anhydride hardeners in a 1:1.2 molar ratio were ground in a mortar at room temperature to obtain a homogeneous mixture. DMID catalyst (molar ratio 0.05 with respect to epoxy monomer) was added and intimately

mixed. Polymerization reaction was studied using DSC. A few milligrams of this mixture was heated from room temperature to 180 °C at a heating rate of 5 °C min<sup>-1</sup>. The remaining mixture was cured in an oven at 150 °C for 1 h.



**Fig. 2.** Molecular representations of reagents used in the resin manufacturing process. Anhydride hardeners: diphenic anhydride (DPA), glutaric anhydride (GA), phthalic anhydride (PA), camphoric anhydride (CA), succinic anhydride (SA), itaconic anhydride (IA). Catalyst: 1,2-dimethylimidazole (DMID).

### 3. Results and discussion

#### 3.1. Optimization of curing protocol by FTIR analysis

The optimization of the epoxy monomer/anhydride hardener molar ratio was carried out on the DGEBA/PA mixture. On the basis of the equal molar functional groups, 1 mol of epoxy monomer is needed for 2 mol of anhydride hardener giving rise to a ring opening polymerization reaction (because the epoxy monomer is tetrafunctional in the absence of labile proton whereas the anhydride hardener is bifunctional) [12,22]. Nevertheless, because of the potential presence of water molecule it can be assumed that the presence of labile protons could reduce the functionality of the epoxy group. Moreover, as mentioned by Dusek [30] and Paramarta and Webster [22], secondary reactions involving only epoxy groups leading to polyetherification also occur. This hypothesis was confirmed here by curing DGEBA only in the presence of DMID catalyst using the protocol mentioned in Section 2. The exothermic peak at 134 °C observed in DSC evidences the polyetherification reaction occurring during the first heating of this mixture. No exothermic peak is recovered during the second heating cycle indicating that there is no more polymerization reaction during the second heating (see Fig. 3). Consequently, it can be assumed that less than 2 mol of anhydride hardener is needed to react with 1 mol of the epoxy monomer.

To optimize the epoxy monomer/anhydride hardener molar ratio (formally equal to 1:2), several mixtures varying from 1:1.5 to 1:0.8 (epoxy monomer/anhydride hardener molar ratio) were polymerized. Results are reported in Table 1. The best molar ratio is determined on the basis of both FT-IR analysis and  $T_g$  values. Indeed the FT-IR spectra

are indicative of the curing reaction. As previously reported [21], the polymerization reaction is highlighted by the appearance of the C=O ester stretching band at about  $1723\text{ cm}^{-1}$ . Concomitantly, the epoxy band at  $915\text{ cm}^{-1}$  (CeO oxirane group stretching) present in the DGEBA spectrum and the anhydride elongation bands at  $1774$  and  $1850\text{ cm}^{-1}$  (C=O) and at  $895\text{ cm}^{-1}$  (CeO) present in the PA spectrum disappear. With the epoxy monomer/anhydride hardener molar ratio of 1:0.8, IR spectrum indicates an epoxy excess ( $\nu_{\text{CeO}} 915\text{ cm}^{-1}$ ), whereas an excess of anhydride remains when using the epoxy monomer/anhydride hardener molar ratio of 1:1.5 ( $\nu_{\text{C=O}} 1774$  and  $1850\text{ cm}^{-1}$ ,  $\nu_{\text{CeO}} 895\text{ cm}^{-1}$ ). With the epoxy monomer/anhydride hardener molar ratio of 1:1.2, the FT-IR spectrum only shows the ester band at  $1723\text{ cm}^{-1}$  (see Fig. 4). Moreover, this molar ratio leads to a high  $T_g$  value of  $110\text{ °C}$  compatible with further applications of the final material. Thus, the best epoxy monomer/anhydride hardener/catalyst molar ratio of 1:1.2:0.05 is used for further work.

#### 3.2. Curing of epoxy monomers with the biosourced anhydride hardeners and DMID catalyst in the molar ratio of 1:1.2:0.05

The curing of epoxy monomers and anhydride hardeners was carried out by heating the mixture of epoxy monomer/anhydride hardener/DMID catalyst. With the aim of application in the composite field, the consistency of the initial mixture has to be taken into account. Depending on the nature of both the epoxy monomer and the anhydride hardener, the initial mixture could be in the solid form or in the form of a waxy paste (WP) or cohesive powder (CP) more or less viscous at room temperature as reported in Table 2. Most of the time, the mixture is in form

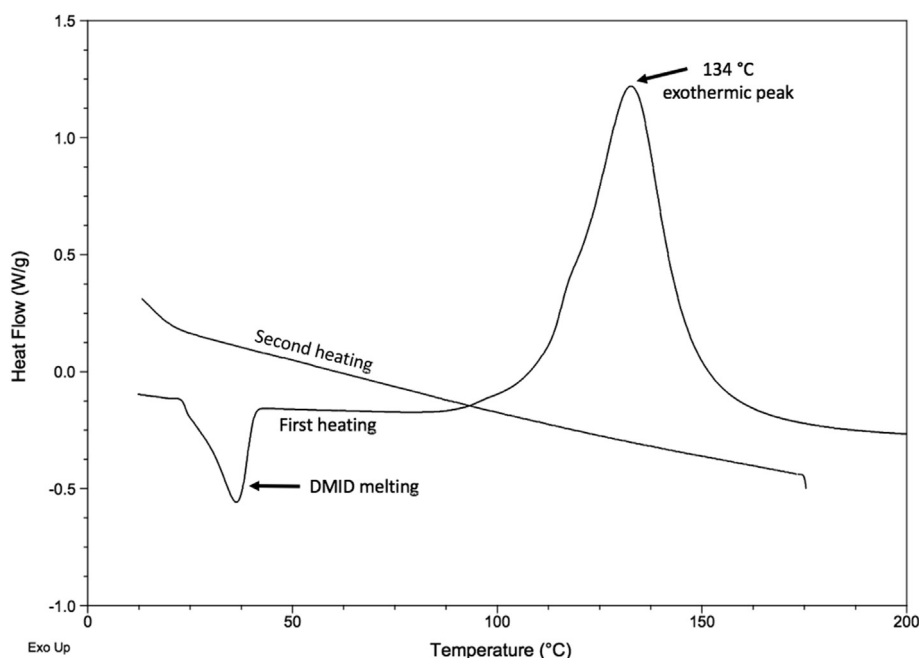


Fig. 3. DSC curves of the curing reaction of DGEBA in the presence of the catalytic amount of DMID.

**Table 1**FT-IR analysis and  $T_g$  values for several epoxy monomer/anhydride hardener/catalyst molar ratios.

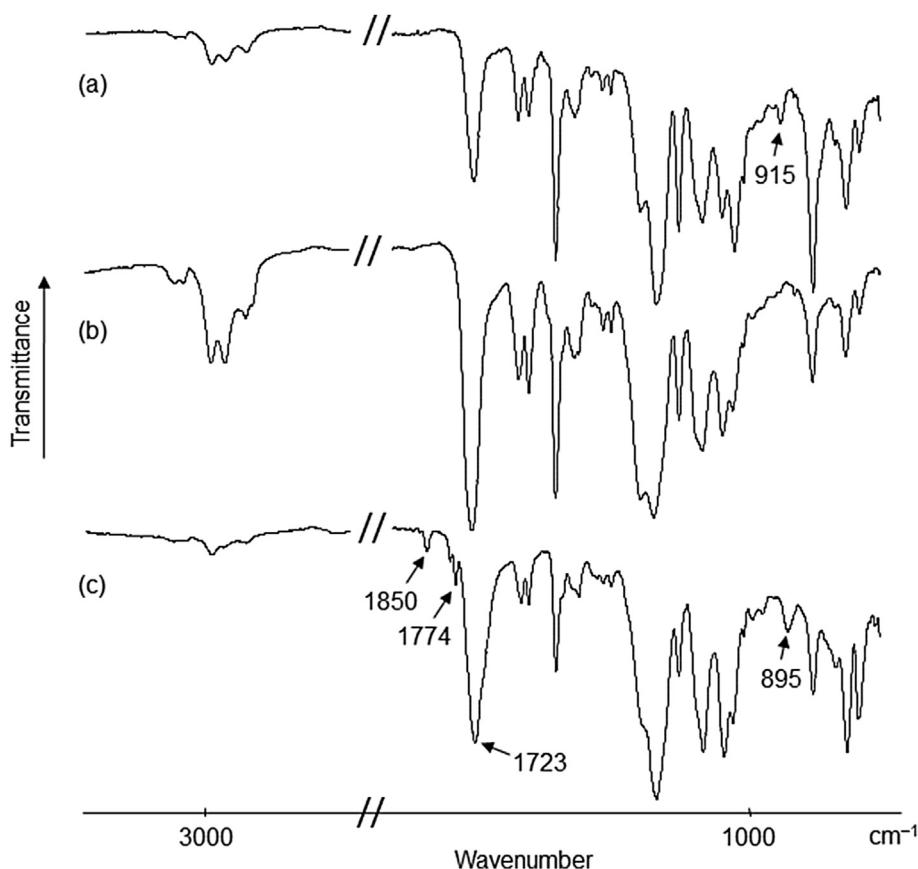
| Molar ratio of monomer/hardener/catalyst | FT-IR                                                                                                                      | $T_g$ (°C) |
|------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|------------|
| 1:0.8:0.05                               | Presence of remaining epoxy ( $915\text{ cm}^{-1}$ ) and resulting ester ( $1723\text{ cm}^{-1}$ )                         | 90         |
| 1:1.2:0.05                               | Only the presence of the resulting ester ( $1723\text{ cm}^{-1}$ )                                                         | 110        |
| 1:1.5:0.05                               | Presence of remaining anhydride ( $895$ , $1774$ and $1850\text{ cm}^{-1}$ ) and resulting ester ( $1723\text{ cm}^{-1}$ ) | 135        |

of WP at room temperature except for the DGE-isoEu for which a CP is often obtained. It is noticeable that all the mixtures become liquid at  $60\text{ °C}$  before the polymerization starts. Furthermore, we verified experimentally by TGA that no physical phenomena inherent to the reactants involved (in particular potential sublimation of acid anhydrides) interfere with the curing process. Namely, no weight losses were recorded in the range of polymerization temperatures. Therefore, all of these experimental data are well compatible with usual composite processing.

Immediately after mixing and for each formulation, polymerization reaction was followed by DSC (heating from  $0$  to  $250\text{ °C}$  at a rate of  $5\text{ °C min}^{-1}$ ). Most of the time one broad exothermic peak is recorded between  $60$  and  $150\text{--}190\text{ °C}$ . However, in some cases, two distinct exothermic peaks are observed. For example, polymerization of DGE-isoEu and IA leads to two peaks starting at

room temperature and ending at  $175\text{ °C}$ . This phenomenon has previously been attributed to possible parallel curing reactions [14,21]. Furthermore, it is worth noting that the exothermic peak can overlap with the end of the endothermic fusion peak of the epoxy monomer or the anhydride hardener compound. In such case, the calculation of the experimental polymerization enthalpy is impossible.

With the final aim of biocomposite manufacturing, attention must be paid to both temperatures: (1) temperature at the beginning of the polymerization and (2) temperature(s) corresponding to the maximum of the polymerization peak(s). Indeed, the polymerization must be avoided during the step of fibre impregnation. Moreover, to prevent damage of fibres, reticulation reaction has to be carried out at less than  $160\text{ °C}$ . Consequently, considering DGE-isoEu monomer, IA hardener is not a good candidate



**Fig. 4.** Infrared spectra attenuated total reflectance of the resulting materials in the epoxy monomer/anhydride hardener/catalyst molar ratio: (a) 1:0.8:0.05, (b) 1:1.2:0.05, (c) 1:1.5:0.05.

**Table 2**

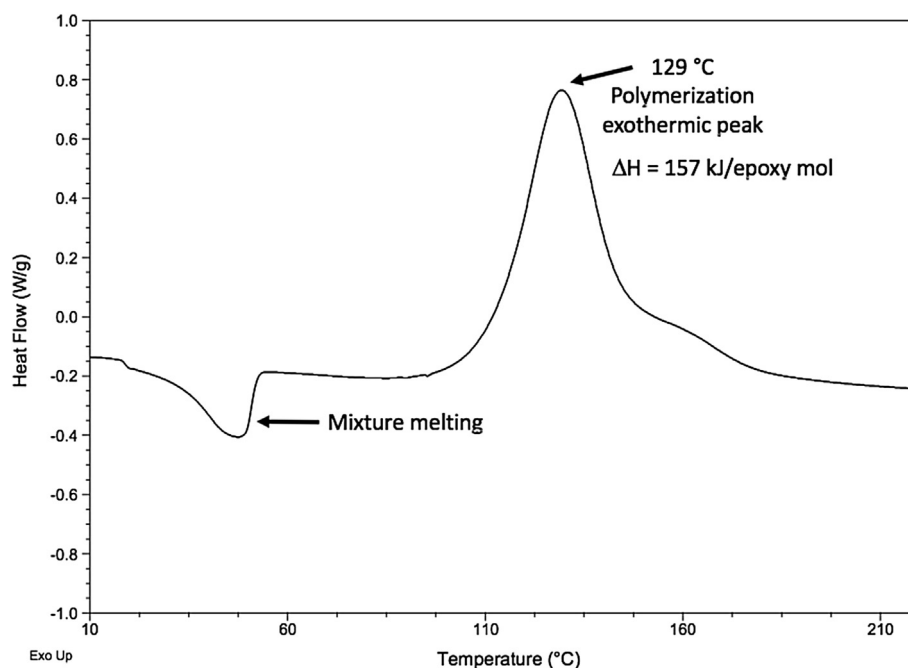
Values of the polymerization enthalpy/temperature and consistency of the different mixtures before polymerization.

|                                               | DPA      | GA       | PA     | CA      | SA     | IA       |
|-----------------------------------------------|----------|----------|--------|---------|--------|----------|
| <b>DGE-isoEu/anhydride hardeners</b>          |          |          |        |         |        |          |
| Mixture consistency                           | CP       | WP       | CP     | CP      | CP     | WP       |
| Polymerization                                |          |          |        |         |        |          |
| Start–end $T$ (°C)                            | 60–150   | 60–160   | 60–190 | 100–190 | 65–150 | 25–175   |
| Peak $T$ (°C)                                 | 108      | 103      | 91     | 129     | 98     | 41; 136  |
| Enthalpy <sup>a</sup> (kJ mol <sup>-1</sup> ) | /        | /        | /      | 157     | /      | /        |
| <b>DGE-Eu/anhydride hardeners</b>             |          |          |        |         |        |          |
| Mixture consistency                           | CP       | WP       | WP     | WP      | WP     | WP       |
| Polymerization                                |          |          |        |         |        |          |
| Start–end $T$ (°C)                            | 50–160   | 60–150   | 70–160 | 80–190  | 75–210 | 50–180   |
| Peak $T$ (°C)                                 | 105; 119 | 132      | 122    | 137     | 125    | 116; 151 |
| Enthalpy <sup>a</sup> (kJ mol <sup>-1</sup> ) | /        | /        | 156    | 131     | 214    | /        |
| <b>DGER/anhydride hardeners</b>               |          |          |        |         |        |          |
| Mixture consistency                           | CP       | WP       | WP     | WP      | WP     | Liquid   |
| Polymerization                                |          |          |        |         |        |          |
| Start–end $T$ (°C)                            | 50–170   | 50–170   | 60–180 | 70–180  | 70–170 | 60–170   |
| Peak $T$ (°C)                                 | 106; 116 | 116; 131 | 112    | 126     | 120    | 109; 122 |
| Enthalpy <sup>a</sup> (kJ mol <sup>-1</sup> ) | /        | /        | /      | /       | /      | /        |
| <b>DGEBA/anhydride hardeners</b>              |          |          |        |         |        |          |
| Mixture consistency                           | CP       | WP       | WP     | WP      | WP     | WP       |
| Polymerization                                |          |          |        |         |        |          |
| Start–end $T$ (°C)                            | 60–170   | 50–210   | 60–180 | 80–200  | 80–150 | 80–160   |
| Peak $T$ (°C)                                 | 121      | 127      | 118    | 133     | 131    | 124      |
| Enthalpy <sup>a</sup> (kJ mol <sup>-1</sup> ) | /        | /        | /      | /       | /      | /        |

<sup>a</sup> The calculation of the exothermic polymerization enthalpy (in kJ mol<sup>-1</sup> of the epoxide group) was not always possible because the exothermic peak is not well dissociated from the endothermic fusion peak of the epoxy monomer/anhydride hardener/DMID catalyst mixture.

because it already reacts at room temperature, whereas CA hardener is the most suitable curing agent. When comparing the polymerization temperatures ( $T$  (°C) of the maximum of the peak), DGE-isoEu monomer appears to be the epoxy resin reacting at the lowest temperature

whatever the hardener. Thus, DGE-isoEu monomer is a well appropriate resin for manufacturing biocomposite reinforced by natural fibres. A typical DSC curve recorded during the curing of DGE-isoEu monomer and CA hardener is reported in Fig. 5.



**Fig. 5.** Typical DSC curve registered during the curing of epoxy monomer with the anhydride hardener according to the procedure described in Section 2 (DGE-isoEu/CA/DMID).

**Table 3**

$T_g$  values of the different epoxy thermosetting polymers determined by DSC.

| Composition of the cured samples | Glass transition temperature ( $T_g$ , °C) |    |     |     |     |    |
|----------------------------------|--------------------------------------------|----|-----|-----|-----|----|
|                                  | DPA                                        | GA | PA  | CA  | SA  | IA |
| DGE-isoEu/hardener               | 123                                        | 78 | 115 | 121 | 94  | 76 |
| DGE-Eu/hardener                  | 126                                        | 65 | 109 | 112 | 79  | 85 |
| DGER/hardener                    | 149                                        | 64 | 124 | 153 | 83  | 74 |
| DGEBA/hardener                   | 152                                        | 90 | 125 | 162 | 101 | 81 |

### 3.3. Characterization of epoxy thermosets

#### 3.3.1. Glass transition temperatures

Glass transition temperatures of the cured samples, determined by DSC, are reported in Table 3. A typical DSC curve showing the  $T_g$  value is reported in Fig. 6 (here, DGE-isoEu/CA/DMID). For all the studied epoxy polymers, no additional exotherms are observed in DSC after the curing schedule indicating that the systems are well cured. The high  $T_g$  values are close to those of the commercial petro-sourced epoxy materials. For instance,  $T_g$  of 100–150 °C are usually obtained when using DGEBA and polyamines [12]. Furthermore, it is notable that these values are very higher than those of the commercial biosourced epoxy materials (Epobiox Sandtech  $T_g$  = 80 °C or Greenpoxy Sicomin  $T_g$  = 85 °C [31]). Thus, these first results are very encouraging for a biocomposite structural application.

$T_g$  reflects both the chemical structure of the polymer chain and the architecture of the polymer as the length and the cross-linking degree of the chain. Thus, as expected, materials obtained using GA, SA or IA hardeners exhibit lower glass transition temperatures. Namely, when these

anhydride cycles are opened during the curing, these hardeners lead to flexible linear carbon chain. Conversely, when anhydride functions of DPA, PA and CA react, bulky and rigid side groups' cycle still remain, leading to materials with higher glass transition temperatures.

Not surprisingly as the chemical structures of DGE-isoEu and DGE-Eu are similar, the glass temperatures of their corresponding materials are close. Therefore, the difference of the epoxide group position (internal position for DGE-isoEu, terminal position for the DGE-Eu) generally inducing a difference in reactivity is not highlighted in this case.

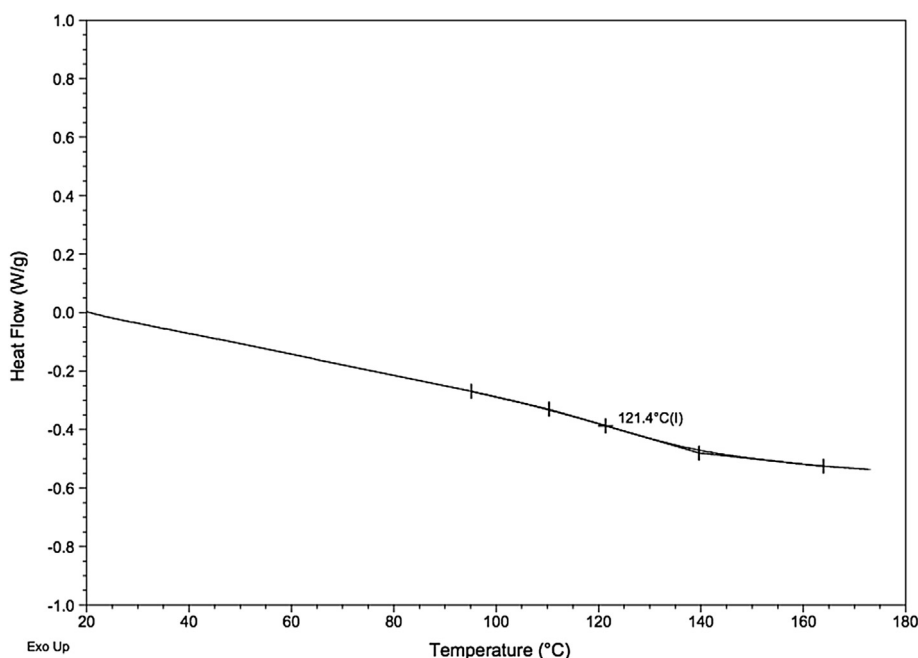
#### 3.3.2. Thermal degradation by TGA

Degradation temperatures of each epoxy thermoset corresponding to 5%, 50% weight loss and maximum degradation rate are labelled as  $T_{5\%}$ ,  $T_{50\%}$  and  $T_{max}$  and reported in Table 4. The polymers have a fairly good thermal stability because they are stable till about 300 °C except when IA is used. In this case  $T_{5\%}$  is recorded at about 270–280 °C.

#### 3.3.3. Nanoindentation measurements

All the samples were submitted successfully to nanoindentation tests except eight materials (DGE-isoEu/DPA, DGE-isoEu/IA, DGE-Eu/DPA, DGE-Eu/GA, DGE-Eu/SA, DGE-Eu/IA, DGER/DPA, and DGEBA/DPA). The heterogeneity and/or porosity of these last materials after polymerization were too high to ensure appropriate and relevant measurements using nanoindentation. It is possible to note that all the materials obtained using the GA hardener were unsatisfactory (Table 5).

An example of the results obtained for each type of indentation tests performed on the DGE-isoEu/SA can be



**Fig. 6.** Determination of the  $T_g$  value (DGE-isoEu/CA/DMID) by DSC according to the procedure described in Section 2.



**Table 4**

Temperatures of thermal degradation ( $T_{5\%}$ ,  $T_{50\%}$  and  $T_{\max}$ ) of the materials resulting from epoxy monomers cured with the different anhydride hardeners.

| Thermoset     | $T_{5\%}$ (°C) | $T_{50\%}$ (°C) | $T_{\max}$ (°C) |
|---------------|----------------|-----------------|-----------------|
| DGE-isoEu/DPA | 332            | 415             | 416             |
| DGE-isoEu/GA  | 328            | 412             | 411             |
| DGE-IsoEu/PA  | 298            | 376             | 379             |
| DGE-isoEu/CA  | 328            | 414             | 426             |
| DGE-isoEu/SA  | 297            | 408             | 409             |
| DGE-isoEu/IA  | 272            | 408             | 410             |
| DGE-Eu/DPA    | 335            | 420             | 420             |
| DGE-Eu/GA     | 355            | 440             | 431             |
| DGE-Eu/PA     | 331            | 395             | 366             |
| DGE-Eu/CA     | 358            | 419             | 414             |
| DGE-Eu/SA     | 331            | 434             | 430             |
| DGE-Eu/IA     | 300            | 427             | 421             |
| DGER/DPA      | 346            | 421             | 420             |
| DGER/GA       | 365            | 440             | 433             |
| DGER/PA       | 321            | 409             | 449             |
| DGER/CA       | 355            | 417             | 417             |
| DGER/SA       | 332            | 430             | 429             |
| DGER/IA       | 272            | 411             | 420             |
| DGEBA/DPA     | 373            | 441             | 429             |
| DGEBA/GA      | 373            | 446             | 441             |
| DGEBA/PA      | 351            | 407             | 395             |
| DGEBA/CA      | 369            | 435             | 426             |
| DGEBA/SA      | 308            | 437             | 433             |
| DGEBA/IA      | 287            | 421             | 428             |

found in Figs. 7–9. The reproducibility can be appreciated in Fig. 9 where three indentation curves are presented and appear almost superimposed. All the figures illustrate the viscous behaviour of the sample. In fact, the response of the sample depends clearly on the values of the constant strain rate  $\dot{h}/h$  (see Fig. 7). Furthermore, the multicycle

indentation curve, shown in Fig. 8, exhibits also open hysteresis on the unloading–reloading segments that are generally characteristic of phase transformation, reversible kink bands or time-dependent behaviour. Finally, the holding time of 1200 s leads to a large amount of material creep, as observed in Fig. 9.

This type of behaviour is observed in all the samples. The strain rate sensitivity calculated assuming a G'Sell and Jonas [32] law allows us to compare the different samples; however, viscoelasticity or viscoplasticity cannot be distinguished.

Table 5 presents the mechanical properties determined using the nanoindentation tests. The materials synthesized from the DGEBA have an instantaneous modulus comprised between 3.36 and 4.73 GPa, a relaxed modulus between 3.02 and 4.24 GPa and a hardness between 183 and 289 MPa. All the materials synthesized from the bio-based polymers have equal or superior properties with an instantaneous modulus comprised between 4.31 and 6.10 GPa, a relaxed modulus between 3.41 and 4.94 GPa and a hardness between 188 and 367 MPa.

The time-dependency is also generally slightly higher for materials synthesized from biobased monomers with a strain rate sensitivity comprised between 0.025 and 0.046 (when compared to 0.026 obtained with DGEBA).

#### 4. Conclusions

DGE-Eu and DGE-isoEu monomers were successfully cured with several biosourced anhydride hardeners affording fully biobased materials. These hardeners have also been tested with the commercial biobased DGER and with the well-known petrosourced DGEBA.

**Table 5**

Properties measured with nanoindentation technique: relaxed ( $E_R$ ) and instantaneous ( $E_I$ ) moduli, hardness ( $H$ ) and strain rate sensitivity ( $m$ ).

| Nanoindentation |          |                                        |                                        |                                      |                                    |
|-----------------|----------|----------------------------------------|----------------------------------------|--------------------------------------|------------------------------------|
| Monomer         | Hardener | $E_R$ (GPa) average standard deviation | $E_I$ (GPa) average standard deviation | $H$ (MPa) average standard deviation | $m$ (°) average standard deviation |
| DGE-isoEu       | DPA      | /                                      | /                                      | /                                    | /                                  |
|                 | GA       | 3.42 ± 0.05                            | 4.26 ± 0.19                            | 234 ± 7                              | 0.032 ± 0.008                      |
|                 | PA       | 4.07 ± 0.01                            | 4.39 ± 0.065                           | 355 ± 5                              | 0.032 ± 0.008                      |
|                 | CA       | 4.06 ± 0.05                            | 4.77 ± 0.12                            | 297 ± 6                              | 0.027 ± 0.003                      |
|                 | SA       | 4.18 ± 0.02                            | 4.95 ± 0.12                            | 285 ± 8                              | 0.046 ± 0.007                      |
|                 | IA       | /                                      | /                                      | /                                    | /                                  |
| DGE-Eu          | DPA      | /                                      | /                                      | /                                    | /                                  |
|                 | GA       | /                                      | /                                      | /                                    | /                                  |
|                 | PA       | 4.67 ± 0.006                           | 5.31 ± 0.104                           | 367 ± 5                              | 0.025 ± 0.006                      |
|                 | CA       | 4.05 ± 0.012                           | 4.68 ± 0.098                           | 301 ± 4                              | 0.025 ± 0.009                      |
|                 | SA       | /                                      | /                                      | /                                    | /                                  |
|                 | IA       | /                                      | /                                      | /                                    | /                                  |
| DGER            | DPA      | /                                      | /                                      | /                                    | /                                  |
|                 | GA       | 3.41 ± 0.11                            | 4.43 ± 0.20                            | 188 ± 12                             | 0.042 ± 0.025                      |
|                 | PA       | 4.91 ± 0.03                            | 5.45 ± 0.14                            | 361 ± 11                             | 0.030 ± 0.007                      |
|                 | CA       | 3.70 ± 0.05                            | 4.31 ± 0.14                            | 238 ± 14                             | 0.029 ± 0.008                      |
|                 | SA       | 4.43 ± 0.00                            | 5.65 ± 0.22                            | 253 ± 7                              | 0.042 ± 0.011                      |
|                 | IA       | 4.94 ± 0.01                            | 6.10 ± 0.18                            | 296 ± 5                              | 0.040 ± 0.009                      |
| DGEBA           | DPA      | /                                      | /                                      | /                                    | /                                  |
|                 | GA       | 3.02 ± 0.03                            | 3.36 ± 0.12                            | 183 ± 10                             | 0.030 ± 0.032                      |
|                 | PA       | 4.24 ± 0.09                            | 4.73 ± 0.16                            | 278 ± 18                             | 0.026 ± 0.014                      |
|                 | CA       | 3.42 ± 0.02                            | 3.50 ± 0.09                            | 220 ± 17                             | 0.026 ± 0.010                      |
|                 | SA       | 3.65 ± 0.01                            | 4.14 ± 0.13                            | 223 ± 10                             | 0.026 ± 0.010                      |
|                 | IA       | 4.21 ± 0.10                            | 4.45 ± 0.09                            | 289 ± 12                             | 0.027 ± 0.013                      |

The quality of the sample surface did not allow satisfactory measurements.



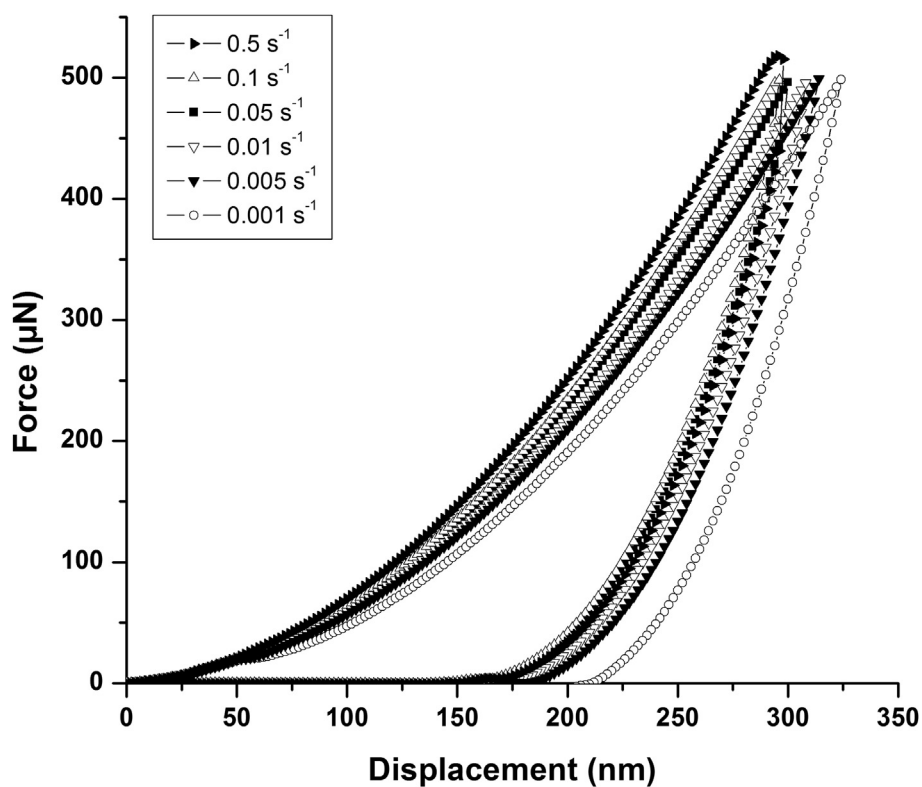


Fig. 7. Load–displacement responses of the DGE-isoEu/SA material obtained at different constant strain rates.

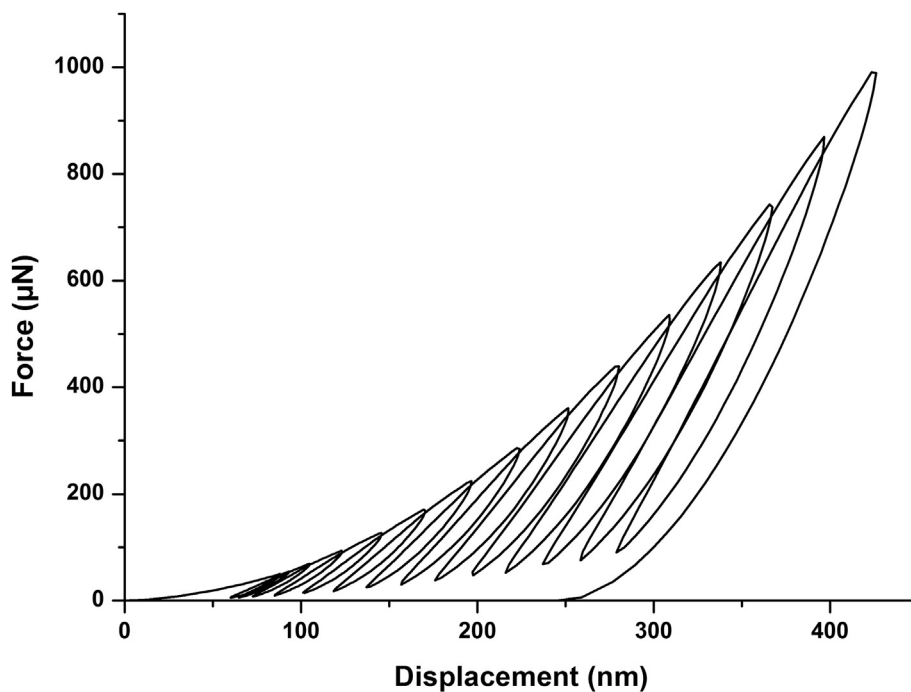


Fig. 8. Load–displacement responses of the DGE-isoEu/SA material during repeated linear load–time indentation ramps with the same rising time (20 s), but different increasing peak loads.

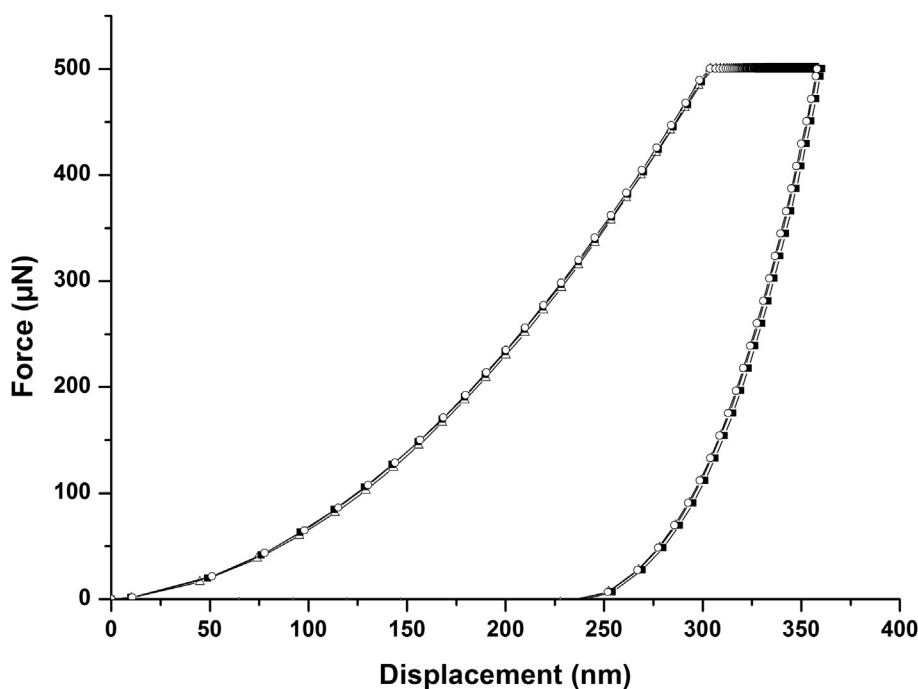


Fig. 9. Load–displacement response obtained under the indenting DGE-isoEu/SA material. A hold period of 1200 s at constant load is applied before unloading.

The biobased thermosetting polymers exhibit satisfactory thermomechanical properties, with  $T_g$  higher than 100 °C and instantaneous modulus ranged from 4 to 6 GPa. Their performances are very close to those of the commercial petrosourced epoxy materials and better than those obtained with the commercial partially biobased epoxy thermosets. The curing of DGE-Eu or DGE-isoEU monomers with DPA, PA, CA hardeners, respectively, leads to the most promising epoxy thermosets.

Polymerization occurs in the range of 90–130 °C. These temperatures are compatible with usual composite processing including natural fibres. This allows the manufacture of fully biobased composite. The study sets up a framework for the impregnation of natural fibres to manufacture fully biobased composites.

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